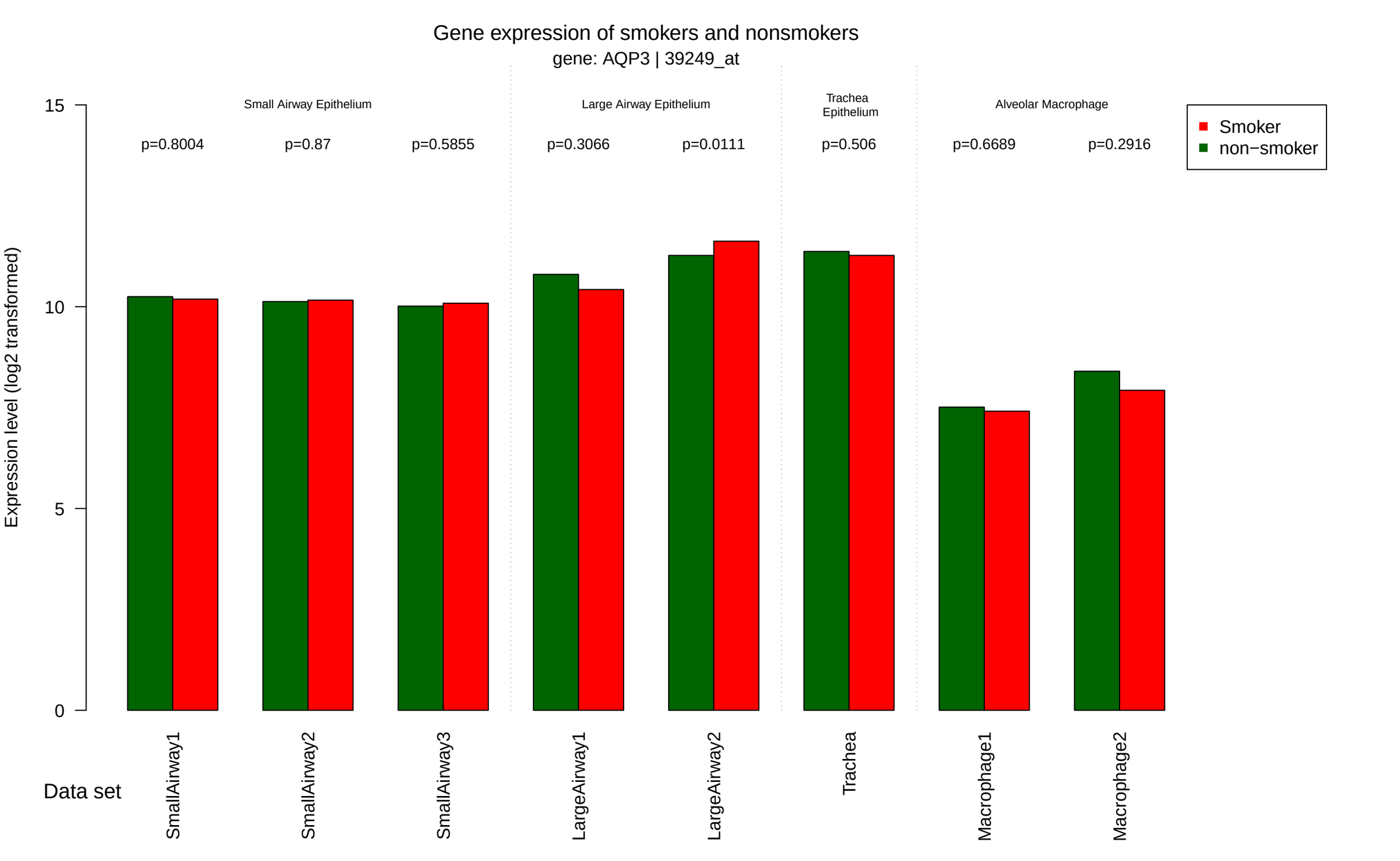
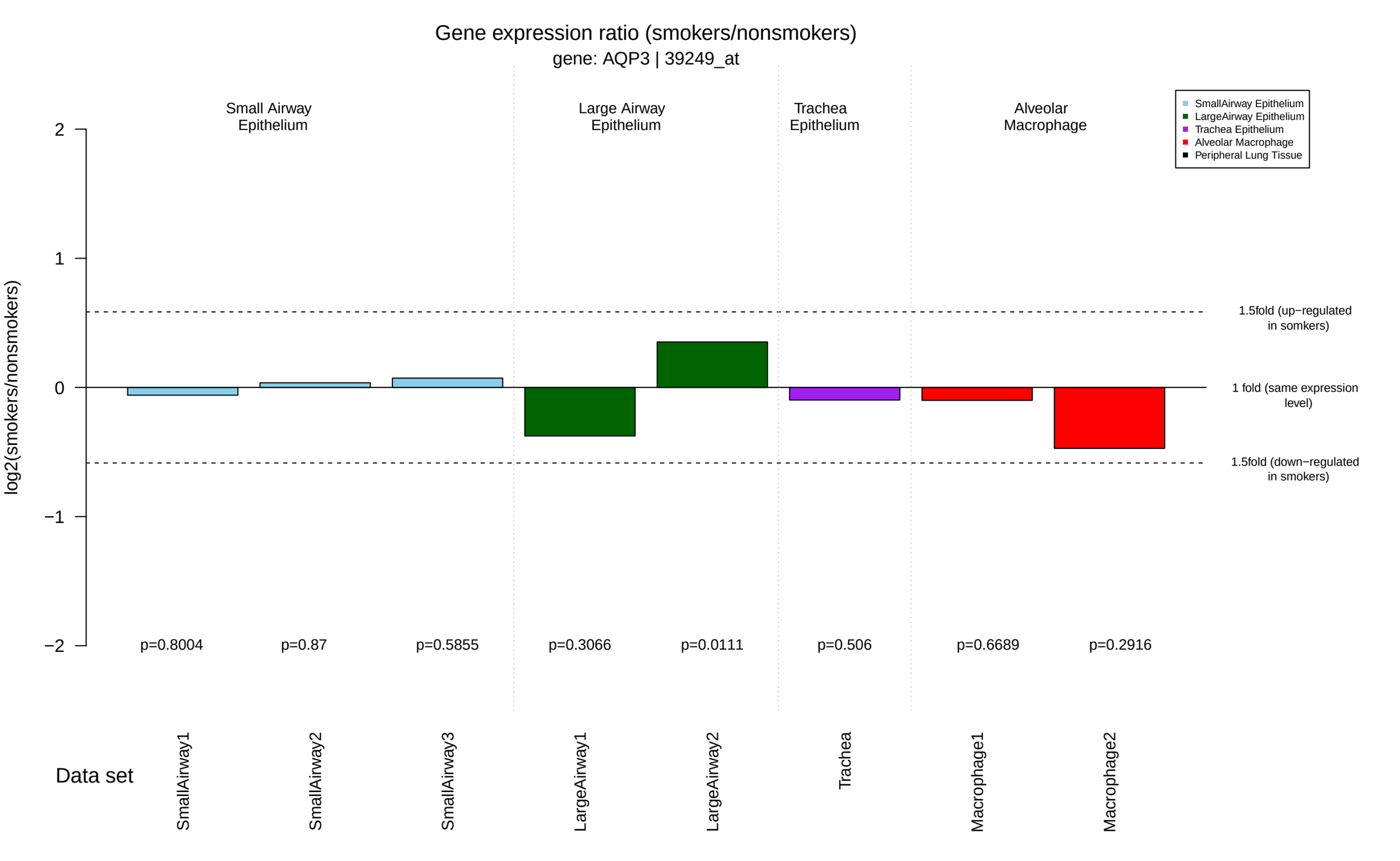
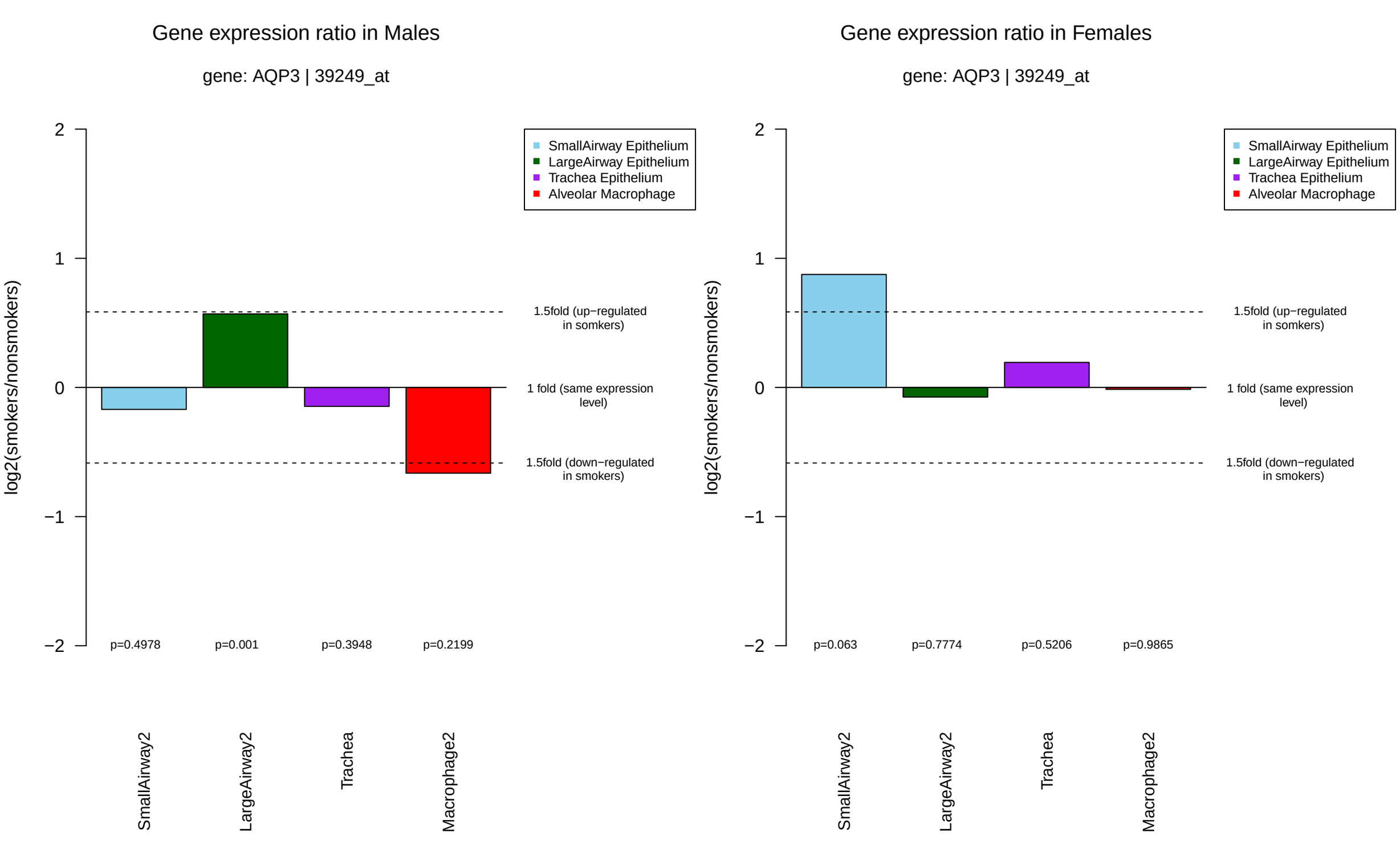
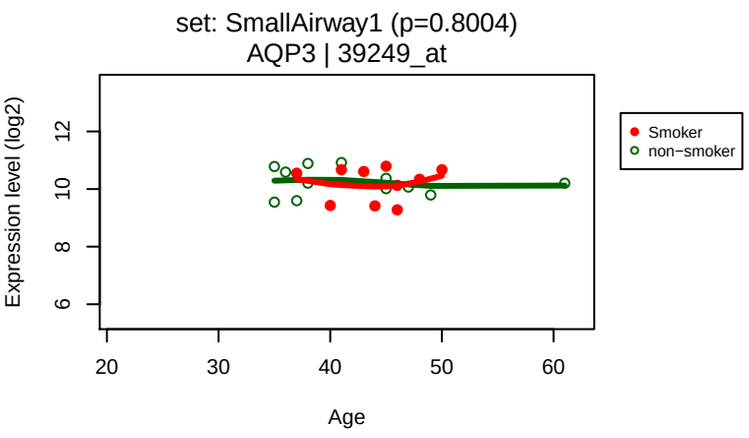
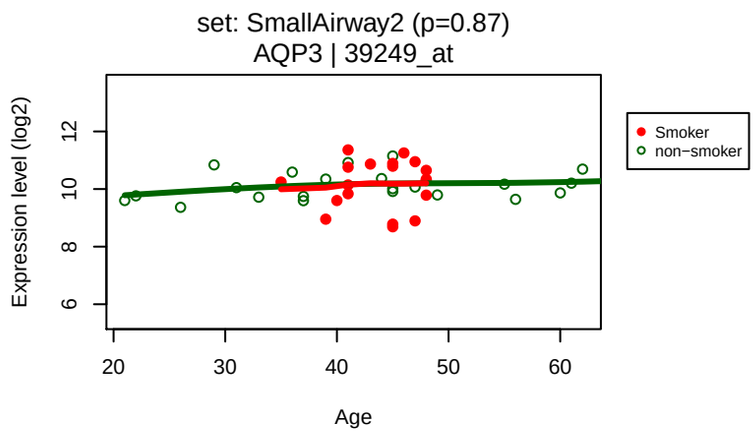
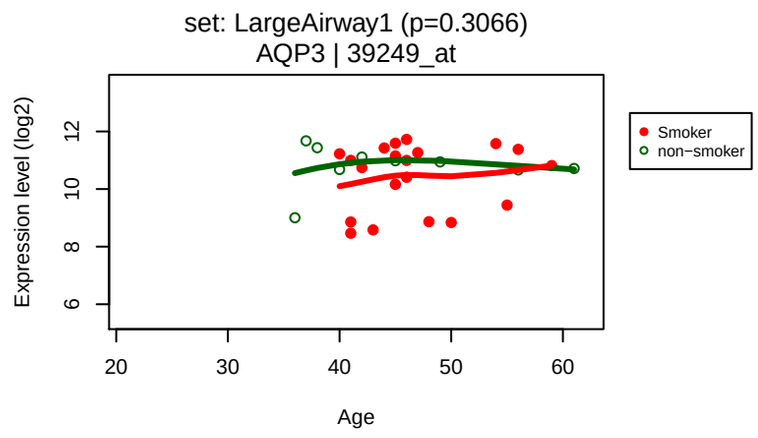

Gene expression of smokers and nonsmokers
gene: AQP3 | 39249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8004
p=0.87
p=0.5855
p=0.3066
p=0.0111
p=0.506
p=0.6689
p=0.2916
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

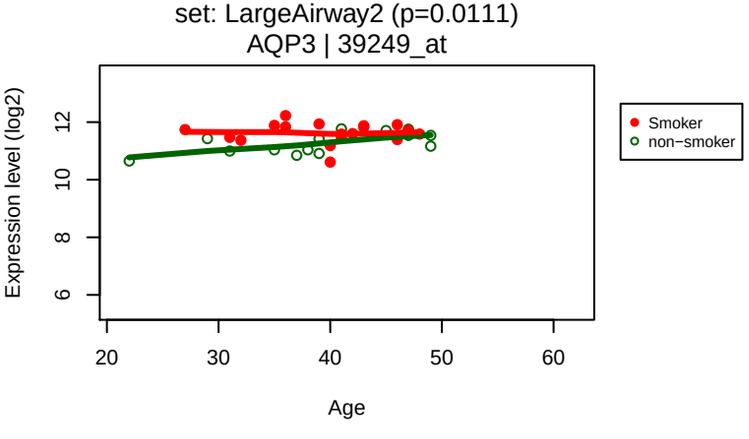

Gene expression ratio (smokers/nonsmokers)
gene: AQP3 | 39249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8004
p=0.87
p=0.5855
p=0.3066
p=0.0111
p=0.506
p=0.6689
p=0.2916
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

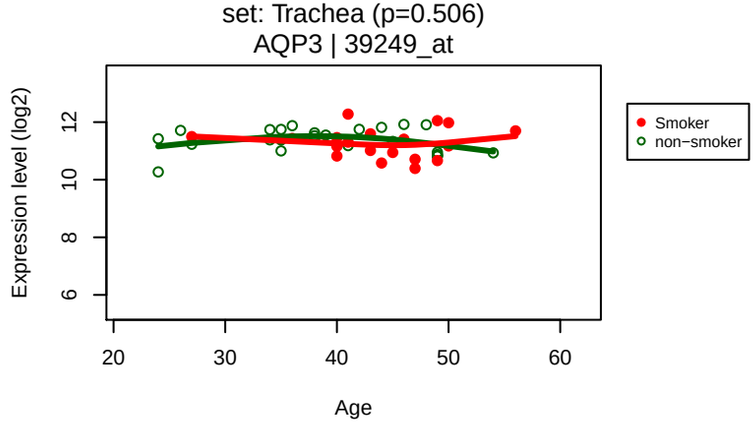

Gene expression ratio in Males
gene: AQP3 | 39249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4978
p=0.001
p=0.3948
p=0.2199
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP3 | 39249_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.063
p=0.7774
p=0.5206
p=0.9865
SmallAirway2
LargeAirway2
Trachea
Macrophage2

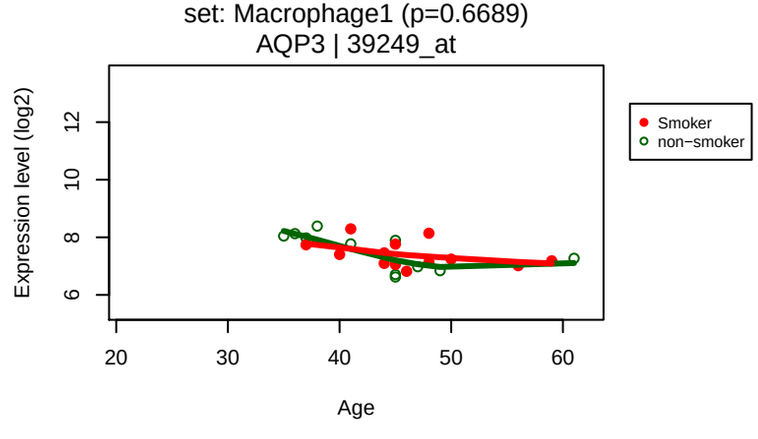

set: SmallAirway1 (p=0.8004)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker

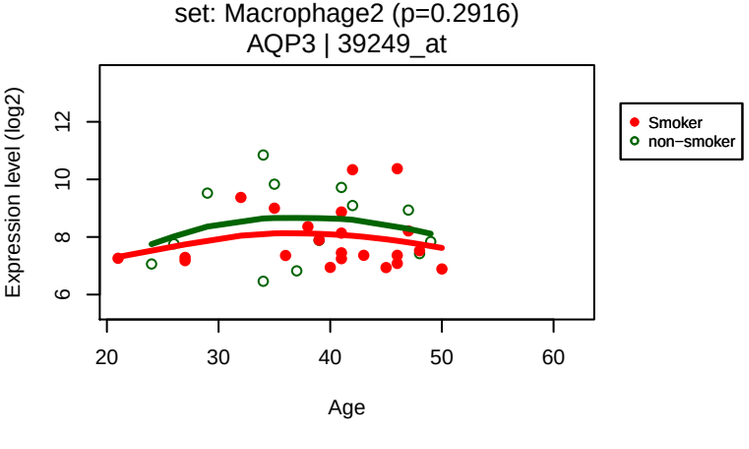

set: SmallAirway2 (p=0.87)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker

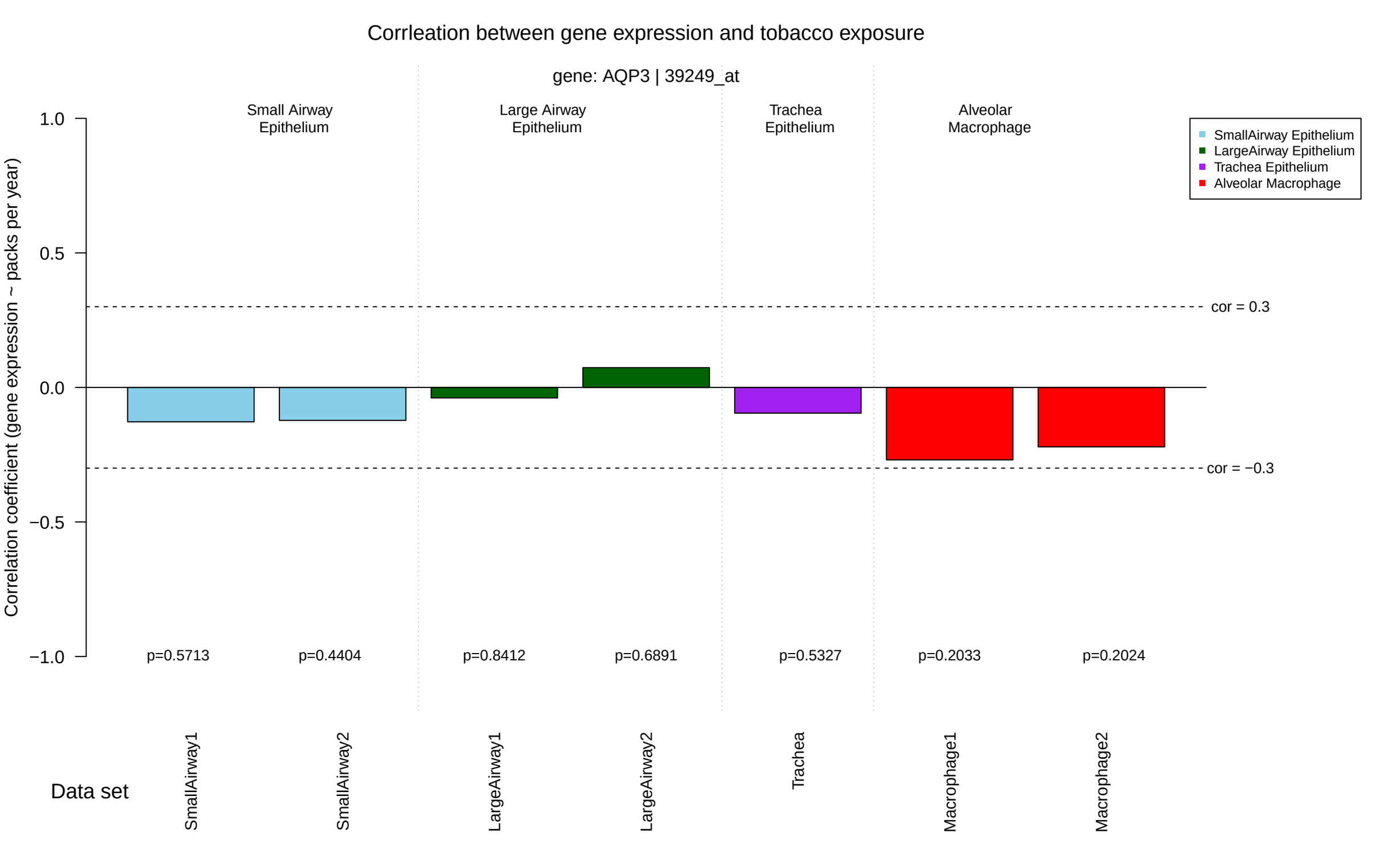

set: LargeAirway1 (p=0.3066)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0111)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.506)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6689)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2916)
AQP3 | 39249_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP3 | 39249_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.5713
p=0.4404
p=0.8412
p=0.6891
p=0.5327
p=0.2033
p=0.2024
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2